

1.5 Install Sysmon Monitoring Service on windows

System Monitor (Sysmon) is a Windows system service and device driver that, once installed on a system, remains resident across system reboots to monitor and log system activity to the Windows event log. It provides detailed information about process creations, network connections, and changes to file creation time.

- The service runs as a **protected process**, thus disallowing a wide range of user mode interactions.
- **Sysmon Download Link** >> <https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>
- stored in Applications and Services Logs/Microsoft/Windows/Sysmon/Operational

Overview of Sysmon Capabilities

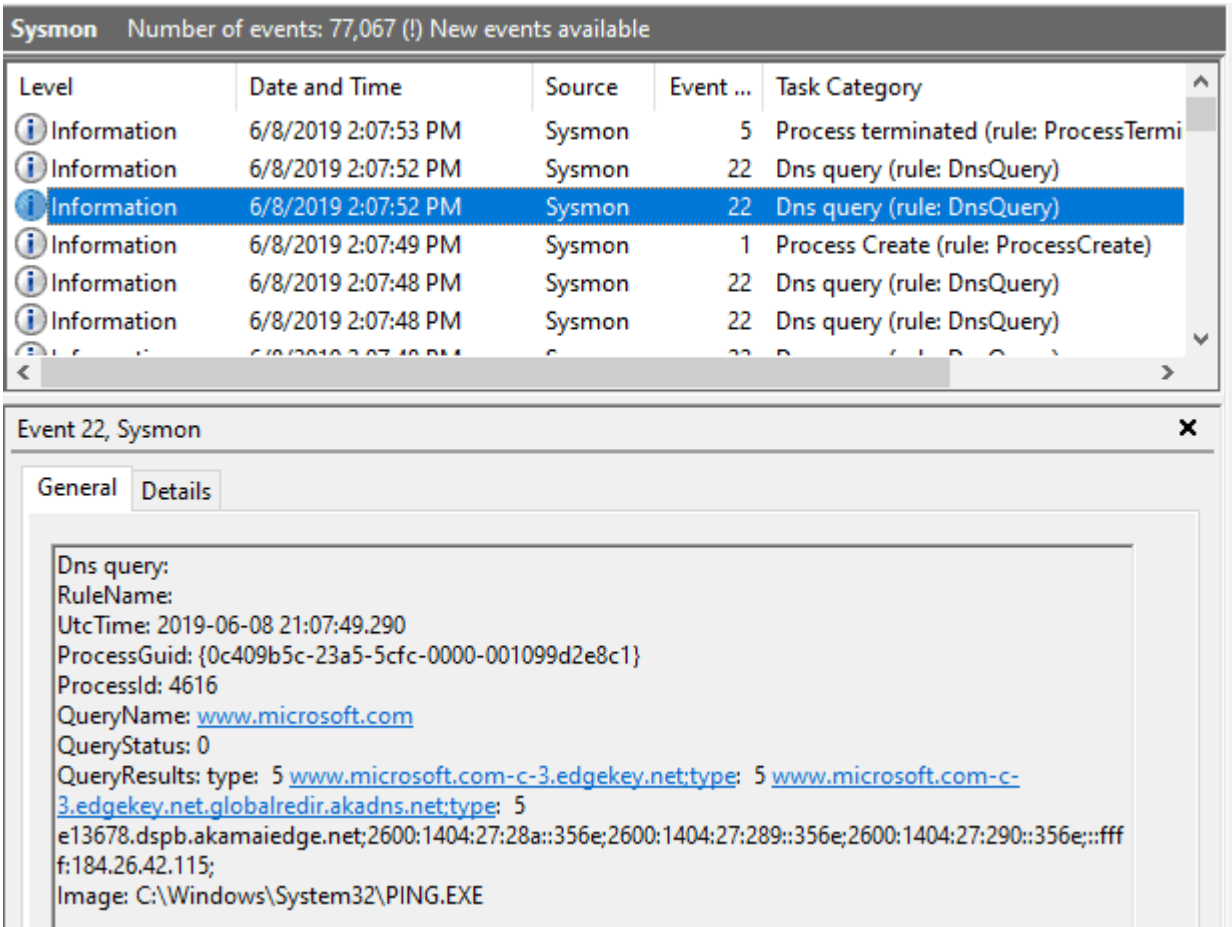
Sysmon includes the following capabilities:

- Logs process creation with full command line for both current and parent processes.
 - Records the hash of process image files using SHA1 (the default), MD5, SHA256 or IMPHASH.
 - Multiple hashes can be used at the same time.
 - Includes a process GUID in process create events to allow for correlation of events even when Windows reuses process IDs.
 - Includes a session GUID in each event to allow correlation of events on same logon session.
 - Logs loading of drivers or DLLs with their signatures and hashes.
 - Logs opens for raw read access of disks and volumes.
 - Optionally logs network connections, including each connection's source process, IP addresses, port numbers, hostnames and port names.
 - Detects changes in file creation time to understand when a file was really created. Modification of file create timestamps is a technique commonly used by malware to cover its tracks.
 - Automatically reload configuration if changed in the registry.
 - Rule filtering to include or exclude certain events dynamically.
 - Generates events from early in the boot process to capture activity made by even sophisticated kernel-mode malware.
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install Sysmon

Windows Command Prompt

```
sysmon -accepteula -i
```



Sysmon Number of events: 77,067 (!) New events available

Level	Date and Time	Source	Event ...	Task Category
Information	6/8/2019 2:07:53 PM	Sysmon	5	Process terminated (rule: ProcessTermi
Information	6/8/2019 2:07:52 PM	Sysmon	22	Dns query (rule: DnsQuery)
Information	6/8/2019 2:07:52 PM	Sysmon	22	Dns query (rule: DnsQuery)
Information	6/8/2019 2:07:49 PM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	6/8/2019 2:07:48 PM	Sysmon	22	Dns query (rule: DnsQuery)
Information	6/8/2019 2:07:48 PM	Sysmon	22	Dns query (rule: DnsQuery)

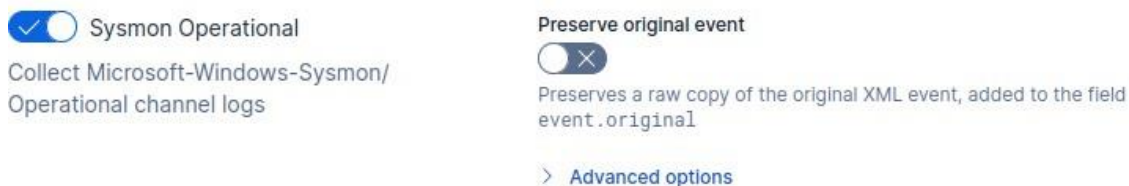
Event 22, Sysmon

General Details

Dns query:
RuleName:
UtcTime: 2019-06-08 21:07:49.290
ProcessGuid: {0c409b5c-23a5-5cfc-0000-001099d2e8c1}
ProcessId: 4616
QueryName: www.microsoft.com
QueryStatus: 0
QueryResults: type: 5 www.microsoft.com-c-3.edgekey.net;type: 5 www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net;type: 5
e13678.dspb.akamaiedge.net;2600:1404:27:28a::356e;2600:1404:27:289::356e;2600:1404:27:290::356e::ffff:184.26.42.115;
Image: C:\Windows\System32\PING.EXE

Display the Sysmon log in Kibana

- 1- Go to Kibana >> fleet >> windows agent >> windows policy
- 2- enable Sysmon Operational button to allow collect logs from Sysmon service



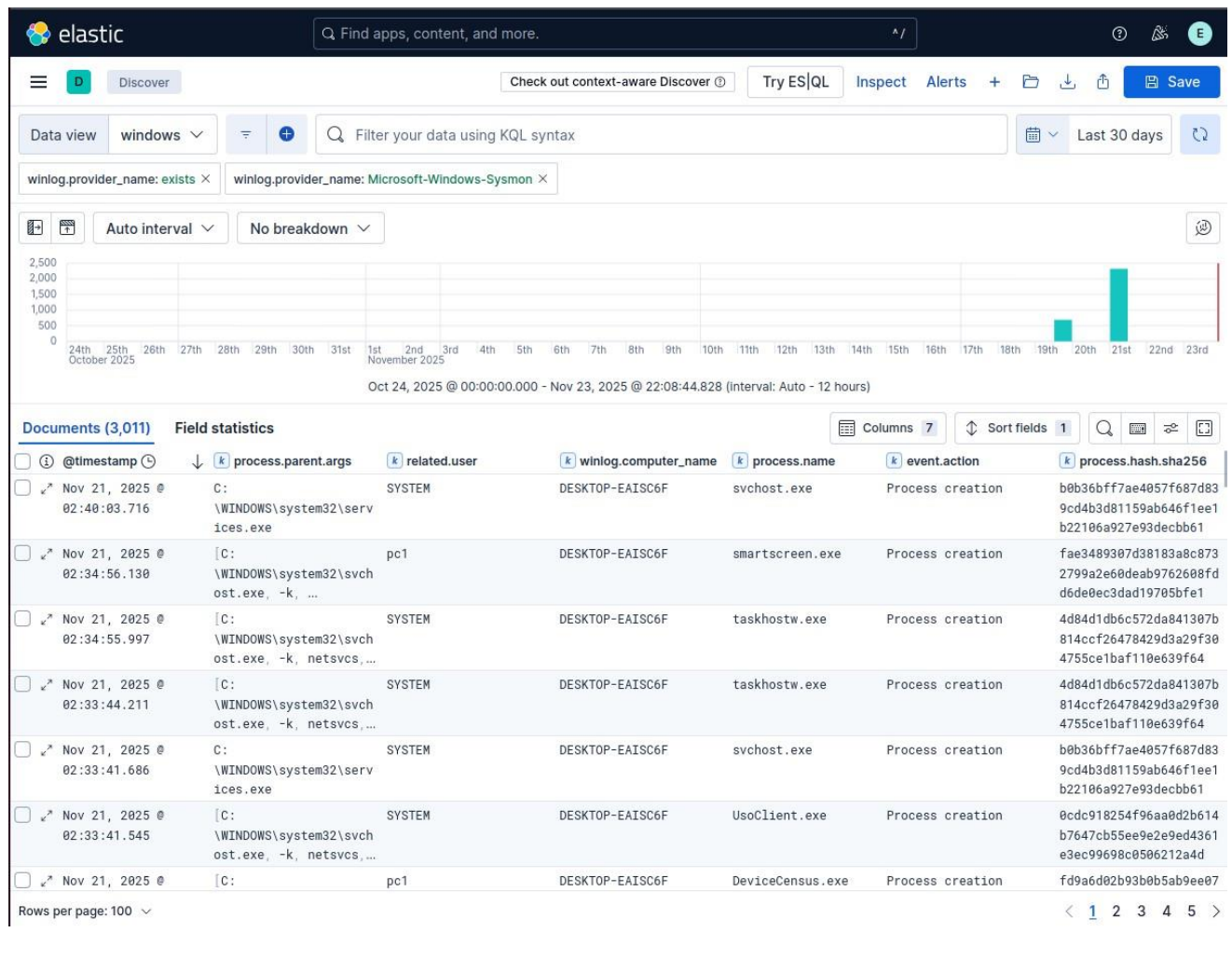
☒ Sysmon Operational

Collect Microsoft-Windows-Sysmon/
Operational channel logs

☐ Preserve original event

Preserves a raw copy of the original XML event, added to the field `event.original`

[Advanced options](#)



Sysmon installed and configured on elastic agent successfully .

